Ruiqian (Rhema) Zhang

Available Summer 2027 | 703-585-5525 | rhema@vt.edu | ruiqian.dev | linkedin.com/in/ruiqianzhang | Blacksburg, VA

Education

Virginia Tech Expected May 2028
B.S. Electrical Engineering; ***GPA: 3.51/4.0*** *Blacksburg, VA*

- **Honors:** Tau Beta Pi Engineering Honor Society; Dean's List with Distinction (Fall 2024, Spring 2025)

Technical Skills

EDA & Circuit Design: Altium Designer, KiCad, EasyEDA, LTspice/SPICE, schematic capture, PCB layout, DRC
Power & Analog: DC-DC buck converters, LDO regulation, current sensing, power distribution
Hardware Testing: Fault isolation, functional verification, test procedure development, oscilloscopes, DMMs
Embedded: C/C++, Python, ATmega328P/32U4, ESP32, CAN, UART, SPI, I2C, ADC, PWM
Tools: MATLAB, Git, DAPLink, J-Link
Spoken Languages: Mandarin Chinese (Fluent), English (Fluent), Thai (Advanced)

Engineering Experience

Undergraduate Researcher Sep. 2026 – Present
Center for Power Electronic Systems, Virginia Tech | Advisor: Dr. Liyan Zhu *Blacksburg, VA*

- Modeling coaxial domino wireless power transfer for gate-driver and sensor loads using circuit and EM simulation.
- Simulating PCB coils and extracting inductance, coupling, resistance, and parasitic capacitance for circuit modeling.

Electrical and Embedded Engineer Sep. 2024 – Present
RoboGrinder at Virginia Tech, Competitive Robotics Engineering Design Team *Blacksburg, VA*

- Designed and simulated a 24 V-to-12 V, 72 W buck converter in LTspice, selecting the switch, diode, inductor, and capacitor for 93.4% efficiency and 0.079% ripple.
- Led reliability testing of 4 competition robots, diagnosing CAN, power, and wiring faults with cross-functional teammates; finished at 2nd place in 3v3 at the 2026 ARC Robotics Championship.
- Created a PCB design training project for new members, each designing and populating 2 LED armor-plate PCBAs.

Electrical Hardware Engineer Sep. 2025 – May 2026
VT Competitive Robotics Organization (VT CRO) *Blacksburg, VA*

- Designed and validated a KiCad PCBA controlling 5 DC motors, 11 servos, and 4 solenoids; verified power rails and PWM for a 17th-place finish at IEEE SoutheastCon 2026.

Projects

12-Key Smart Locker | *ATmega328P, Embedded C, ADC, PWM, Power Management* Feb. 2026 – May 2026

- Built a standalone 9 V ATmega328P controller with MCP1702 LDO regulation to 5 V and 50 Hz servo PWM, validating operation with a DMM and oscilloscope.
- Detected and recovered from servo obstructions within 1 second by sampling current through a 1 Ω shunt resistor, applying four-sample ADC averaging and timing filters, and reversing the servo to its previous position.

STC89C52RC Microcontroller Development Board | *EasyEDA, PCB layout, Soldering* Feb. 2025 – Mar. 2025

- Owned end-to-end development of a custom 2-layer 51-pin STC89C52RC microcontroller development board by designing the schematic and PCB layout in EasyEDA, selecting components, developing the BOM, and sourcing parts for assembly.
- Designed PCB layout with calculated trace widths, ground copper pours, and oscillator keepouts to reduce noise coupling; hand-soldered all components.

MePad V1: Programmable USB Keypad | *EasyEDA, ATmega32U4, QMK* Jan. 2026 – May 2026

- Owned end-to-end development of a programmable USB keypad, integrating 9 mechanical keys, 2 rotary encoders for scrolling, and USB connectivity through a custom ATmega32U4 PCB.
- Developed QMK firmware in C with VIA remapping and integrated PCB into a custom 3D-printed Fusion 360 enclosure.

Leadership Experience

Co-Founder and Treasurer Sep. 2026 – Present
IEEE Power & Energy Society Student Chapter, Virginia Tech *Blacksburg, VA*

- Co-founded the chapter with a 5-member team and established financial operations, including banking, event budgets, and student government funding.

Co-Founder and Advisor Aug. 2026 – Present
ICSHacks Club, International Community School Bangkok *Bangkok, Thailand*

- Co-founded and advised a 9-member team organizing Thailand's first student-run high-school hackathon, overseeing workshops, sponsorships, marketing, and school outreach.